

Linked Lists

A linked list stores nodes connected by next pointers. The head pointer identifies the first node. An empty list has a null head.

Insertion at the beginning

To insert at the beginning of a linked list, first connect the new node to the previous head. Then update the head pointer to the new node. This preserves the existing chain.

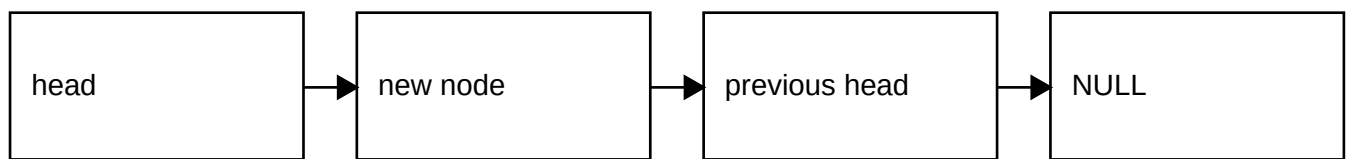


Figure 1: connect new node to previous head, then head to new node.

The conceptual mistake

If the new node's next pointer is set to NULL before updating the head, the old nodes become unreachable. The first insertion can pass while insertion into a non-empty list fails.

Reassessment: count the nodes

To count the nodes, start at head, increment the count once for each visited node, and follow next until NULL. An empty list has zero nodes.